Chat Application

Client-Server Architecture Using Python Socket Programming

Department of Computer Science & Engineering — Bachelor of Technology, Computer Networks (2025-26)

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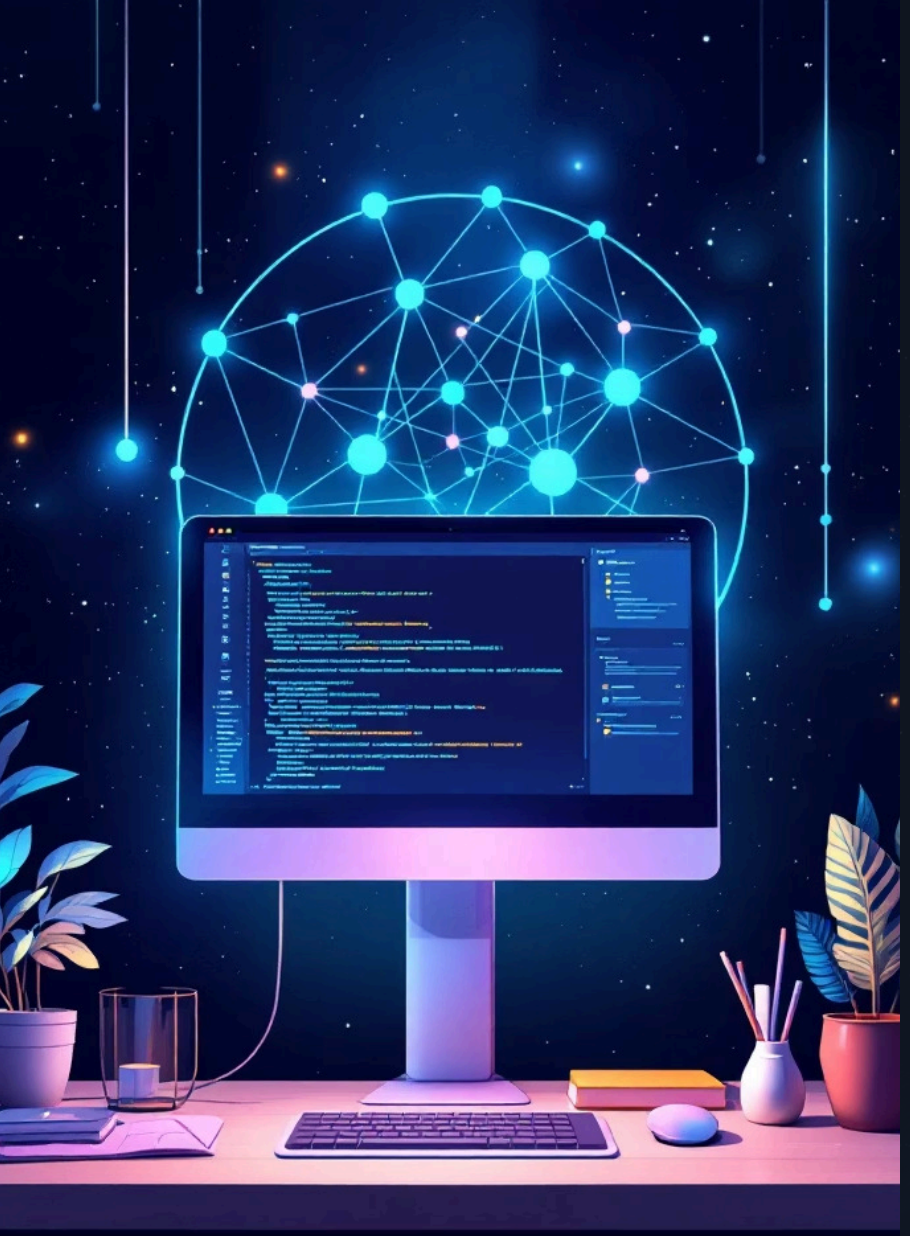

Project Overview

Core Technology

Python 3 socket and threading modules for TCP-based real-time messaging

Network Simulation

Cisco Packet Tracer 8.x for topology visualization and packet-level analysis

A practical demonstration of Client-Server architecture, TCP/IP protocols, and multi-threaded programming for Computer Networks course (4th Semester).

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Key Objectives

Implement Chat System

Build working chat application using Python 3 socket programming

Demonstrate Architecture

Showcase Client-Server model through practical implementation

Handle Multiple Clients

Manage simultaneous connections using threading module

Simulate Network Topology

Verify connectivity with ping tests and PDU packet simulation

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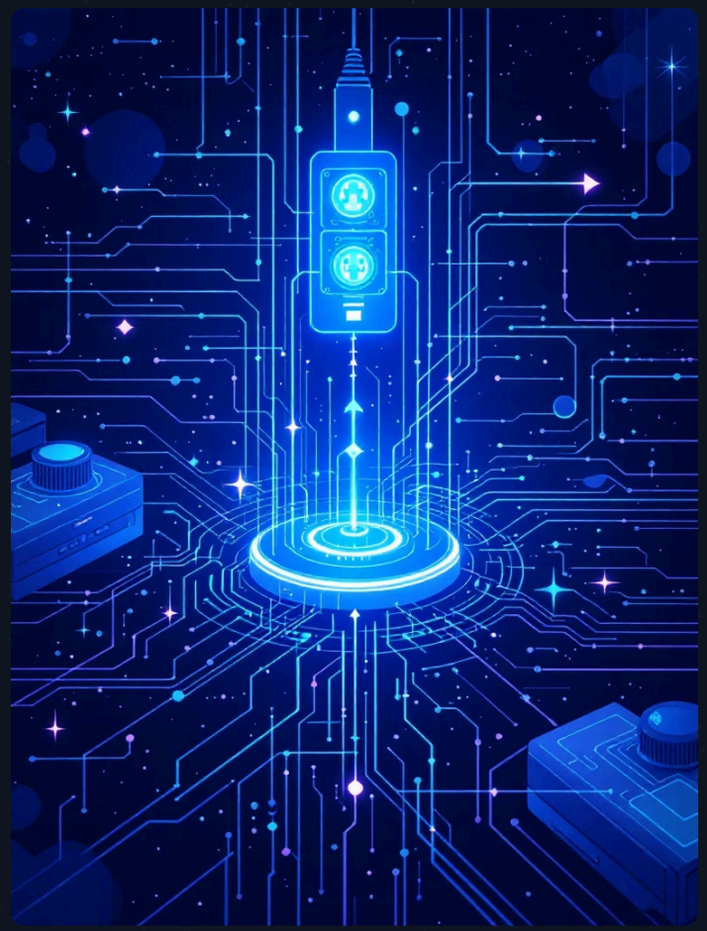
Socket Programming Fundamentals

What is a Socket?

A software abstraction providing an endpoint for communication between processes over a network. Identified by IP address (machine) and port number (application).

TCP Protocol

Connection-oriented protocol at Transport Layer (Layer 4) ensuring reliable, ordered, error-checked delivery through three-way handshake and acknowledgments.



Port 12345 used for server listening

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Client-Server Architecture

Server

Central hub running continuously, accepting connections on port 12345

Client

Initiates connection, provides nickname, sends/receives messages

Broadcast

Server forwards messages to all connected clients simultaneously

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Multi-Threading Implementation

Concurrent Client Handling

Each client connection spawns a dedicated thread using `threading.Thread()`, allowing server to handle multiple clients simultaneously without blocking.

When Thread 1 waits for Alice's message, Thread 2 processes Bob's messages independently.

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OSI Model Layers

Application (7)

Chat logic — send/receive messages

Presentation (6)

`encode('utf-8')`/`decode('utf-8')`

Session (5)

`socket.connect()`/`accept()`

Transport (4)

`socket.SOCK_STREAM` (TCP), port 12345

Network (3)

IP address 127.0.0.1

Data Link (2)

Handled by OS, Ethernet frame via Switch0

Physical (1)

Network card/loopback, Copper cable in PT

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Network Topology

Cisco Packet Tracer simulation with four PCs connected via 2960 Switch

Device	IP Address	Subnet Mask	Role
PC-Server	192.168.1.1	255.255.255.0	Runs server.py
PC-Alice	192.168.1.2	255.255.255.0	Client 1
PC-Bob	192.168.1.3	255.255.255.0	Client 2
PC-Charlie	192.168.1.4	255.255.255.0	Client 3
Switch0 (2960)	N/A	N/A	Layer 2 device

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Implementation Results

Successful Testing

Server handled multiple clients simultaneously without delay or interference

Packet Simulation

PDU details confirmed TCP protocol at Layer 4 with destination port 12345

Connectivity Verified

All PCs pinged each other with 0% packet loss

Server Output

```
Server running on 127.0.0.1:12345
Connection from ('127.0.0.1', 50421)
Nickname: Alice
Connection from ('127.0.0.1', 50422)
Nickname: Bob
```

Client Output

```
Enter your nickname: Alice
Connected to the server!
Alice joined the chat!
Bob joined the chat!
```

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Conclusion & Future Enhancements

Project Success

Successfully demonstrated socket programming, Client-Server architecture, TCP/IP communication, and multi-threading using only standard Python libraries.

GUI Interface

tkinter-based graphical interface

Authentication

Username/password verification

Encryption

AES/RSA message security

File Transfer

Support for document sharing

Private Rooms

Dedicated chat channels

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